

Biomedical Equipment Triage Policy

Kenya Medical Training College — Capstone reference policy, September 2026

Purpose

This policy defines how a reported biomedical equipment maintenance issue is classified by urgency, routed to a responsible team, and documented when required information is missing. It draws on the principles set out in WHO's medical equipment maintenance programme guidance (inspection, preventive maintenance, and corrective maintenance), adapted here into facility-specific triage rules with defined response windows.

1. Urgency Levels

Every report is assigned exactly one urgency level. The level is determined by the observed situation described in the report, not by the equipment type alone.

Level	Definition	Response window
Critical	Equipment failure or fault that directly affects a patient who is currently dependent on that equipment, or that otherwise directly affects patient functioning.	Immediate escalation to specialized team
High	Equipment fault with clear potential to affect patient safety or care if not addressed promptly, even if no adverse clinical consequences have occurred yet.	Within 15-30 minutes
Medium	Equipment fault that affects normal operation or reliability but does not present an immediate safety concern within the current working shift	Within the current working shift
Low	Cosmetic, administrative, or scheduled maintenance matters with no effect on equipment performance	Routine maintenance schedule

2. Precedence when the equipment register and the report disagree

The equipment register's criticality field is a static planning attribute and may lag behind the equipment's actual clinical use at the time of the report. When the register's criticality rating and the report's clinical context point to different urgency levels, the classification follows the observed clinical context in the report, not the register field. The register's criticality rating should still be recorded and referenced in the reasoning, since a mismatch between the two is itself worth flagging back to the equipment inventory owner.

3. Team Routing

The first question for any report is whether the equipment is a registered biomedical equipment asset. That question, not the specific fault type, is what separates Biomedical Engineering from Facilities:

Is it a registered biomedical equipment asset? → **YES:** Biomedical Engineering owns it, regardless of whether the fault is internal, external/cosmetic, performance-related, or routine maintenance. → **NO:** Is it fixed facility infrastructure? If yes, Facilities. If it is not a registered biomedical equipment asset and is not fixed facility infrastructure, apply the specialized routing rules below for ICT, Nursing/Supply Chain, or Vendor Escalation as applicable.

- **Biomedical Engineering** — all equipment listed on the biomedical equipment register, including equipment malfunction, calibration drift, alarm faults, and physical damage affecting the device's internal or external physical condition, regardless of whether the device is currently connected to a patient. This includes portable medical devices such as suction machines and oxygen concentrators.
- **Facilities** — fixed building infrastructure and facility systems, including electrical supply, medical-gas lines, wall suction outlets, and other infrastructure not itself listed as a biomedical equipment asset. Facilities does not assume ownership of a registered portable medical device merely because it is not currently patient-connected.
- **ICT** — connectivity faults, network-dependent data transmission issues, and software/interface faults not tied to the device's mechanical or electrical function.
- **Nursing / Supply Chain** — consumables, accessories, and user-replaceable parts (tubing, cables, probes) where no device fault has been confirmed.
- **Vendor Escalation** — faults requiring manufacturer-level calibration, parts, or diagnostics beyond the facility's in-house biomedical engineering scope (for example, laboratory analyzer calibration).

Revision note (Run 3): This section was clarified after Run 2 produced two textually defensible readings for a portable device with external-only damage and no confirmed internal fault. The boundary is now ownership-based (registered asset vs. fixed infrastructure) rather than fault-type-based, to remove that ambiguity.

4. Handling Incomplete or Ambiguous Information

A report may lack the detail needed to confidently classify urgency, issue type, or team. In that case:

- Do not infer or invent missing details to reach a definitive classification.
- Assign the most severe urgency level that is clearly supportable by the information actually present, then explicitly flag which additional information is required to confirm or downgrade it.

- List the specific missing fields or facts by name (for example: "equipment status not specified", "clinical context not provided") rather than a general statement that information is missing.
- Assign a lower confidence score in proportion to how much of the classification rests on assumption rather than stated fact.

5. Prior Reports on the Same Equipment

Where more than one report references the same `equipment_id`, the classification must take the prior report's content into account. If an earlier report identified a fault and there is no confirmation on file that the fault was resolved, a new report on the same unit should be treated as a possible continuation or recurrence of that fault, and this reasoning should be stated explicitly rather than treating the new report as an unrelated first-time event.

6. Required Output Fields

For every report, the classification output must include: equipment type, issue type, urgency level, assigned team, a confidence score, a short reason for the decision referencing the specific evidence used, and — where applicable — an explicit list of missing information that limited the classification.

Confidence score: The confidence score must be expressed as a decimal from 0.00 to 1.00, where 1.00 represents high confidence and lower values indicate greater uncertainty.